The Standard Trade Model and External Economies of Scale

ECON 308 – International Economic Relations

Fall 2025

Problem 1 – Terms of Trade and Welfare (Standard Trade Model)

A country produces **food (F)** and **manufactures (M)**. Its PPF is:

$$F^2 + M^2 = 400.$$

The world relative price is $P_M/P_F = 2$.

(a) Determine the country's production point that maximizes the value of output.

(b) Suppose consumers prefer $M = 0.5F$. Find the consumption point.

(c) Identify whether the country exports or imports manufactures.

(d) Explain how an increase in P_M/P_F affects the country's welfare.

Problem 2 – Export- and Import-Biased Growth

Country A exports steel and imports textiles. Initially, the world relative price is $P_S/P_T = 1$.

(a) Using a standard trade diagram (PPF + indifference curves), show the effects of **export-biased growth** in Country A on:

- The production point,
- The relative supply of steel, and
- The world relative price.

(b) Discuss qualitatively whether this growth improves or worsens Country A's terms of trade.

(c) Repeat for **import-biased growth** and explain the welfare implications.